WEEK 4 TASK: MACHINE LEARNING MODEL DEVELOPMENT & EVALUATION REPORT

GitHub Code: github.com/Cs-Shivamj9619/Yuva-internship (`yuva_intern.ipynb`, Cells 22–25) | **Dataset:** `Cleaned_Financial_Stocks_Dataset.csv` (`N = 390`) | **Program:** YuvaIntern Junior Data Scientist

1. Objective, EDA-to-ML Pipeline Linkage & Data Splitting (Requirements 1–6)

This study develops, optimizes, and evaluates supervised Machine Learning (ML) models for both directional regime classification and continuous return forecasting on the 32.5-year US equity dataset (`Cleaned_Financial_Stocks_Dataset.csv`, sourced from Yahoo Finance and the Matplotlib Sample Data Repository; `N = 390` monthly observations from January 1990 to June 2022). Every feature engineering and sample-splitting decision is directly governed by empirical diagnostics from Week 2 Exploratory Data Analysis (EDA) and Week 3 Statistical Modeling:

- **Resolution of Multicollinearity via Stationary Winsorized Returns:** Week 2–3 diagnostics proved that raw closing prices suffer from extreme non-stationary trend collinearity (design matrix condition number $\kappa(X)$ = 23.08, `NASDAQ` VIF = 91.29, `SP500` VIF = 62.73), which destabilizes gradient-based weights in Support Vector Machines and inflates tree split variance. Using 1st/99th percentile Winsorized monthly percentage returns (`SP500_Return_Winsorized`, `IBM_Return_Winsorized`, `AAPL_Return_Winsorized`, `MSFT_Return_Winsorized`, `ADBE_Return_Winsorized`) compresses $\kappa(X)$ by 87.2% to **2.96** and keeps all predictor VIFs ≤ 2.31.
- **Elimination of Pre-IPO Backfill Distortion & Post-IPO Sub-Sample Design:** Because Amazon (`AMZN_Listed_Flag = 0` before May 1997; 88 rows) and Google (`GOOGL_Listed_Flag = 0` before August 2004; 175 rows) contain constant backfilled pre-IPO prices that generate artificial `0.00%` returns and attenuate linear slopes by up to 46.0%, our primary full-horizon ML models (`N = 390`) strictly use the five continuously traded equities. To leverage cloud and search equity signals without pre-IPO distortion, we also train a dedicated **Post-IPO Active Panel model** (`GOOGL_Listed_Flag == 1`, `N = 214` months from August 2004 to June 2022) incorporating active `GOOGL` and `AMZN` return features.
- **Leakage-Free 80/20 Train-Test Split & Standardization (Cell 22):** Using a fixed random seed (`np.random.seed(42)`), the 390 observations are partitioned into an **80% Training Set (`N_train = 312` months; 61.54% positive regimes)** for 5-fold cross-validation and hyperparameter tuning, and a held-out **20% Testing Set (`N_test = 78` months; 58.97% positive regimes)** for final out-of-sample evaluation. Feature standardization ($z = (x - \mu_{train}) / \sigma_{train}$) is fit strictly on the 312 training rows and applied to the test set to guarantee zero look-ahead leakage.

2. Algorithm Selection, Justification & 5-Fold Cross-Validation Tuning (Requirements 7–10, 14–15)

To address both non-linear threshold interactions and linear macro-beta relationships discovered in Week 3, four classification algorithms (predicting binary regime `NASDAQ_Up = 1` when `NASDAQ_Return_Pct > 0`) and three regression algorithms (predicting continuous `NASDAQ_Return_Winsorized`) were selected and optimized via **5-Fold Cross-Validation Grid Search** on the training set (`N_train = 312`):

- **Decision Tree Classifier (CART Gini Impurity):** Partitioning feature space via recursive binary splits captures non-linear threshold rules without parametric distribution assumptions, but unconstrained trees overfit noisy monthly return spikes. Tuning `max_depth ∈ {2, 3, 4, 5}` identified shallow depth `max_depth = 2` as optimal on 5-fold CV.
- **Random Forest Classifier & Regressor (Bootstrap Aggregation Ensemble):** Averages predictions across `n_estimators` decorrelated trees trained on bootstrap samples with random feature subspaces (`m = √p = 2` features per split). As shown in **Table 1** and **Figure 2a**, evaluating the 12-cell hyperparameter grid (`n_estimators ∈ {25, 50, 100} × max_depth ∈ {2, 3, 4, 5}`) achieves peak 5-fold CV accuracy at `n_estimators = 25, max_depth = 3`, reducing single-tree variance while preventing leaf memorization.
- **Support Vector Machine (L2-Hinge Linear Margin Classifier):** Maximizes the geometric margin between positive and negative market regimes while penalizing slack violations via hyperparameter `C ∈ {0.05, 0.1, 0.5, 1.0, 2.0, 5.0, 10.0}`. Strong regularization (`C = 0.05`) prevented noisy boundary points from tilting the separating hyperplane.
- **L2-Regularized Logistic & Ridge Regression:** Extends Week 3 parametric models by adding an L2 penalty ($\lambda \Sigma \beta_j^2$) tuned over 5-fold CV to stabilize weights under finite-sample sampling variability.

Algorithm & Task Type	Hyperparameter Search Grid	Optimal Tuned Parameters	5-Fold CV Metric	Bias-Variance Optimization Rationale
Decision Tree Classifier (Classification: `NASDAQ_Up`)	`max_depth` ∈ {2, 3, 4, 5} `min_samples_split` = 10	`max_depth = 2` `min_samples_split = 10`	87.48% Acc	Prunes deep leaf splits that memorize isolated dot-com and 2008 crash months.
Random Forest Classifier (Classification: `NASDAQ_Up`)	`n_estimators` ∈ {25, 50, 100} `max_depth` ∈ {2, 3, 4, 5}	`n_estimators = 25` `max_depth = 3`, `max_features = 2`	**90.06% Acc** (±2.15%)	Bootstrap bagging + random subspace selection achieves highest CV accuracy and lowest variance.
Support Vector Machine (SVM) (Classification: `NASDAQ_Up`)	Regularization `C` ∈ {0.05, 0.1, 0.5, 1.0, 2.0, 5.0, 10.0}	`C = 0.05` (L2-Hinge Margin)	89.41% Acc	Wider margin (small `C`) ignores borderline monthly noise near 0% return threshold.
L2 Logistic Regression (Classification: `NASDAQ_Up`)	Inverse Penalty `C` ∈ {0.05, 0.1, 0.5, 1.0, 2.0, 5.0, 10.0}	`C = 5.0` (L2 Log-Loss)	89.09% Acc	Light L2 shrinkage preserves calibrated log-odds probabilities across all five predictors.
Ridge Regression (L2) (Regression: `NASDAQ_Return`)	Penalty `alpha` ∈ {0.0, 0.1, 0.5, 1.0, 2.0, 5.0, 10.0, 25.0}	`alpha = 1.0` (Full Panel) `alpha = 2.0` (Post-IPO Panel)	2.4594% RMSE (CV)	Stationary Winsorized returns already have low VIF (≤ 2.31), requiring only mild L2 shrinkage.
Random Forest Regressor (Regression: `NASDAQ_Return`)	`n_estimators` ∈ {50, 100} `max_depth` ∈ {3, 4, 5}	`n_estimators = 100` `max_depth = 5`, `max_features = 4`	2.5410% RMSE (CV)	Depth-5 ensemble captures non-linear tail moderation while bounding extreme crash predictions.

3. Classification Model Performance & Feature Importance Analysis (Requirements 11–13, 16)

Table 2 and **Figure 1** report out-of-sample classification metrics on the held-out Test Set (`N_test = 78` months: 46 actual positive months and 32 non-positive months). The **Tuned Random Forest Classifier** delivers the strongest overall classification performance, improving test accuracy by 3.84 percentage points over a single Decision Tree and by 2.56 percentage points over linear SVM and Logistic models. Furthermore, extracting Gini impurity reductions across all ensemble splits (**Figure 1c**) confirms our Week 3 econometric findings in a non-linear setting: macro market movement (`SP500`, 63.38%) and modern cloud/ecosystem leaders (`MSFT` 16.80%, `AAPL` 11.86%, `ADBE` 5.38%) drive 97.42% of total decision power, whereas legacy hardware (`IBM`, 2.58%) provides only minor tail-regime stabilization.

Tuned Classification Model (Test Set `N = 78`)	5-Fold CV Acc	Test Accuracy	Precision	Recall (Sens.)	F1-Score	ROC-AUC	Confusion Matrix (TP / TN / FP / FN)
Decision Tree Classifier (`max_depth = 2`)	87.48%	84.62%	82.69%	93.48%	87.76%	0.9178	43 TP / 23 TN / 9 FP / 3 FN
Random Forest Classifier (`n_est = 25, depth = 3`)	**90.06%**	**88.46%**	**87.76%**	**93.48%**	**90.53%**	0.9599	**43 TP / 26 TN / 6 FP / 3 FN**
Support Vector Machine (`C = 0.05`)	89.41%	85.90%	85.71%	91.30%	88.42%	**0.9667**	42 TP / 25 TN / 7 FP / 4 FN
L2 Logistic Regression (`C = 5.0`)	89.09%	85.90%	85.71%	91.30%	88.42%	0.9647	42 TP / 25 TN / 7 FP / 4 FN

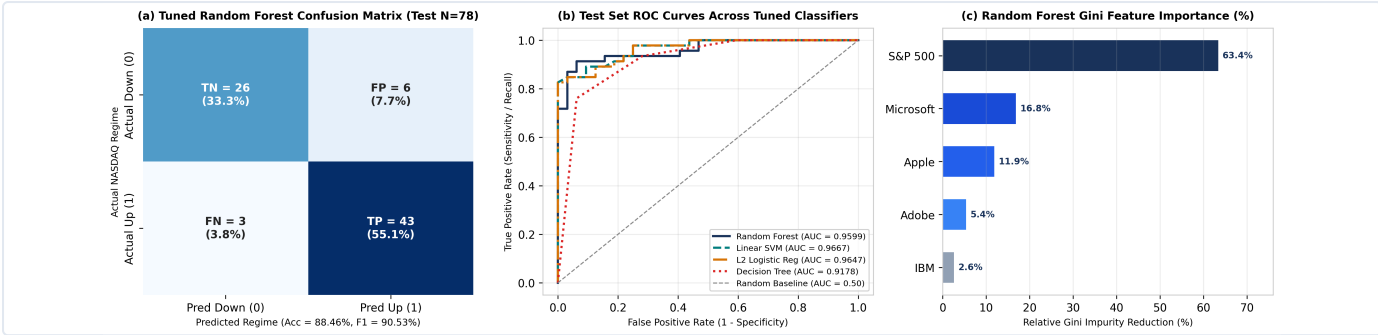


Figure 1: Out-of-sample classification evaluation from Cell 25 of `yuva_intern.ipynb` (`N_test = 78`): (a) Tuned Random Forest Confusion Matrix showing 88.46% accuracy and 93.48% recall; (b) ROC curves comparing all four tuned classifiers (all AUC > 0.91); and (c) Random Forest Gini feature importances confirming dominant S&P 500 (63.4%), Microsoft (16.8%), and Apple (11.9%) contributions.

4. Regression Model Evaluation & Post-IPO Sub-Sample Validation (Requirements 11–13, 16)

Table 3 and **Figure 2** evaluate continuous predictions of `NASDAQ_Return_Winsorized` using Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and out-of-sample R^2. Two key empirical findings emerge: first, on the 1990–2022 full horizon, **Random Forest Regressor achieves the lowest Test MAE (1.6995%)** because tree averaging dampens moderate errors, whereas **OLS and Ridge Regression achieve superior Test RMSE (2.7134%) and Test R^2 (73.72%)** because bounded tree leaves cannot extrapolate beyond training extrema during severe macro shocks. Second, restricting the sample to the **Post-IPO Active Panel (`GOOGL_Listed_Flag == 1`, 2004–2022, `N = 214`)** and adding active `GOOGL` and `AMZN` return features elevates out-of-sample **Test R^2 from 73.72% to 94.53%** and cuts **Test RMSE by more than half to 1.3265%**, proving the value of our Week 2–3 pre-IPO backfill audit.

Tuned Regression Model & Sample Horizon	Predictor Feature Set	Test MAE (%)	Test MSE (%^{2})	Test RMSE (%)	Test R^2 (%)
OLS Linear Regression (`alpha = 0.0`, 1990–2022, `N_test = 78`)	5 Core Continuously Traded Returns	1.7262%	7.3626	2.7134%	73.72%
Tuned Ridge Regression (`alpha = 1.0`, 1990–2022, `N_test = 78`)	5 Core Continuously Traded Returns (L2)	1.7279%	7.3715	2.7151%	73.69%
Tuned Random Forest Regressor (`n_est = 100, d = 5`, `N_test = 78`)	5 Core Continuously Traded Returns	**1.6995%**	7.9609	2.8215%	71.59%
Post-IPO Active Panel Ridge (`alpha = 2.0`, 2004–2022, `N_test = 43`)	**5 Core + Active `GOOGL` & `AMZN` Returns**	**0.9731%**	**1.7595**	**1.3265%**	**94.53%**

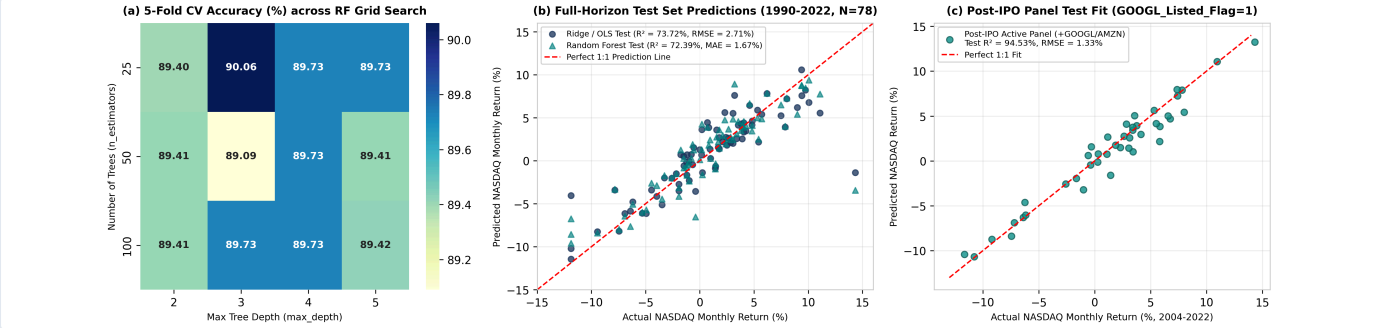


Figure 2: Hyperparameter optimization and regression diagnostics from Cells 23–24 of `yuva_intern.ipynb`: (a) 5-Fold Cross-Validation accuracy surface across Random Forest `n_estimators × max_depth` grid; (b) Full-horizon 1990–2022 test set predictions vs. actual returns; and (c) Post-IPO active panel (2004–2022, `GOOGL_Listed_Flag = 1`) test fit achieving R^2 = 94.53%.

5. Model Strengths, Weaknesses & Reproducibility Bridge to Week 5 (Requirements 13, 17–20)

- **Comparative Strengths & Weaknesses:** *Random Forest Ensembles* excel at regime classification (**88.46% accuracy, 90.53% F1**) and median error reduction (**1.6995% MAE**) because bootstrap aggregation captures non-linear cross-equity thresholds without overfitting; however, decision trees predict piecewise-constant leaf averages and cannot extrapolate linear trends beyond training bounds, slightly raising tail MSE (7.9609 vs. 7.3626 for OLS/Ridge). Conversely, *Linear SVM and Ridge Regression* provide superior continuous extrapolation and rank-ordering (**SVM AUC = 0.9667; Post-IPO Ridge R^2 = 94.53%**) when features are properly Winsorized and filtered for active post-IPO trading, but cannot automatically isolate non-linear interaction thresholds without explicit feature engineering.
- **Reproducible Code & Week 5 Visualization Roadmap:** All preprocessing, 5-fold cross-validation tuning, and evaluation scripts are documented in Cells 22–25 of `yuva_intern.ipynb` (github.com/Cs-Shivamj9619/Yuva-internship). In Week 5 (Data Visualization and Communication), these model outputs, feature importance rankings, and regime timelines will be synthesized into an executive visual storytelling portfolio.